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Preventing hamstring injuries in football through enhanced exercise and RTP strategies

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WHAT DID I DO?

This thesis consisted of seven publications on the prevention of hamstring injuries in male football, targeting primary prevention and return-to-play (RTP) after hamstring injuries.

WHY DID I DO IT?

Hamstring injuries is the number 1 muscle injury in football, and injury rates have been increasing consistently over the last decade.[1] Furthermore, hamstring injuries are notorious for their high recurrence rates, leading to prolonged rehabilitation, personal suffering for the player and decreased team performance.

HOW DID I DO IT?

The thesis kicks off with a systematic review on the effect of general exercise programmes for the prevention of football injuries. The first study was a randomised controlled trial investigating the preventive effect of the Nordic hamstring exercise (NHE) (figure 1) in more than 500 male amateur football players.[2] In the same study, the role of hamstring flexibility in injury prevention was analysed.[3]

For the second study I performed another systematic review, focusing on the definition of and medical criteria for RTP after hamstring injuries.[4] Then, a worldwide Delphi study with experts selected by the FIFA Medical Centres of Excellence network was conducted to discuss RTP after hamstring injuries in football.[5]

WHAT DID I FIND?

The first study showed that incorporating the NHE programme in regular training led to a threefold to fourfold reduction of hamstring injury risk in amateur football.[2] However, the NHE programme did not influence hamstring injury severity.[2] Hamstring flexibility tests—for which normative values for a football population were provided—should not be used to identify players at risk for hamstring injuries.[4]

The second study resulted in the RTP model for hamstring injuries in football (figure 2).[5] The RTP decision, for example, the clinician's criteria-based judgement to medically clear a player for full availability for match selection and/or full training, should always be a multidisciplinary decision. Absence of pain, psychological readiness, similar hamstring flexibility and performance on field tests should be assessed for the RTP decision, whereas MRI was not advised to guide RTP clearance.

WHAT IS THE MOST IMPORTANT CLINICAL IMPACT?

Nordics work for hamstring injury prevention in an amateur football setting![3] Solid evidence for its effectiveness is provided, which is important for medical staff members and players, and for coaches as fewer injuries have been associated with higher league rankings.[6]

Furthermore, my research concluded that hamstring flexibility scores were not able to identify players at risk for future hamstring injuries.[3]

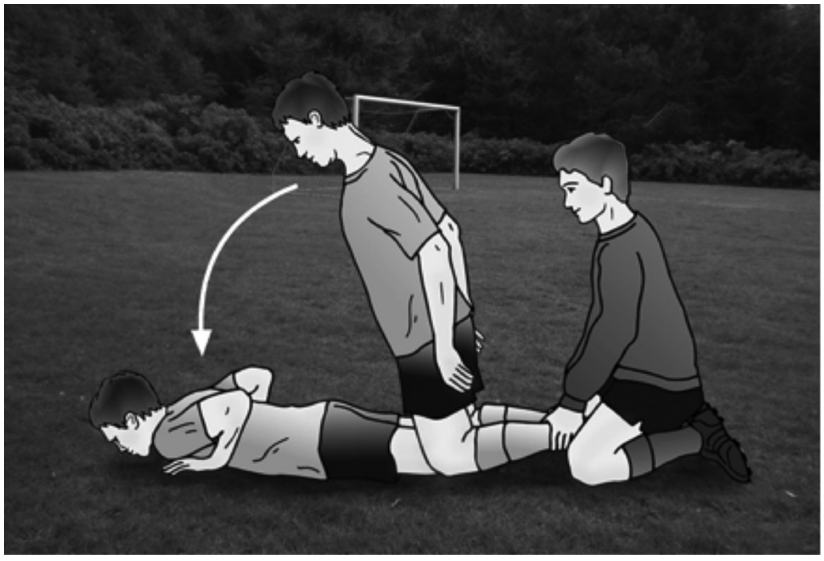

Figure 1 Nordic hamstring exercise.

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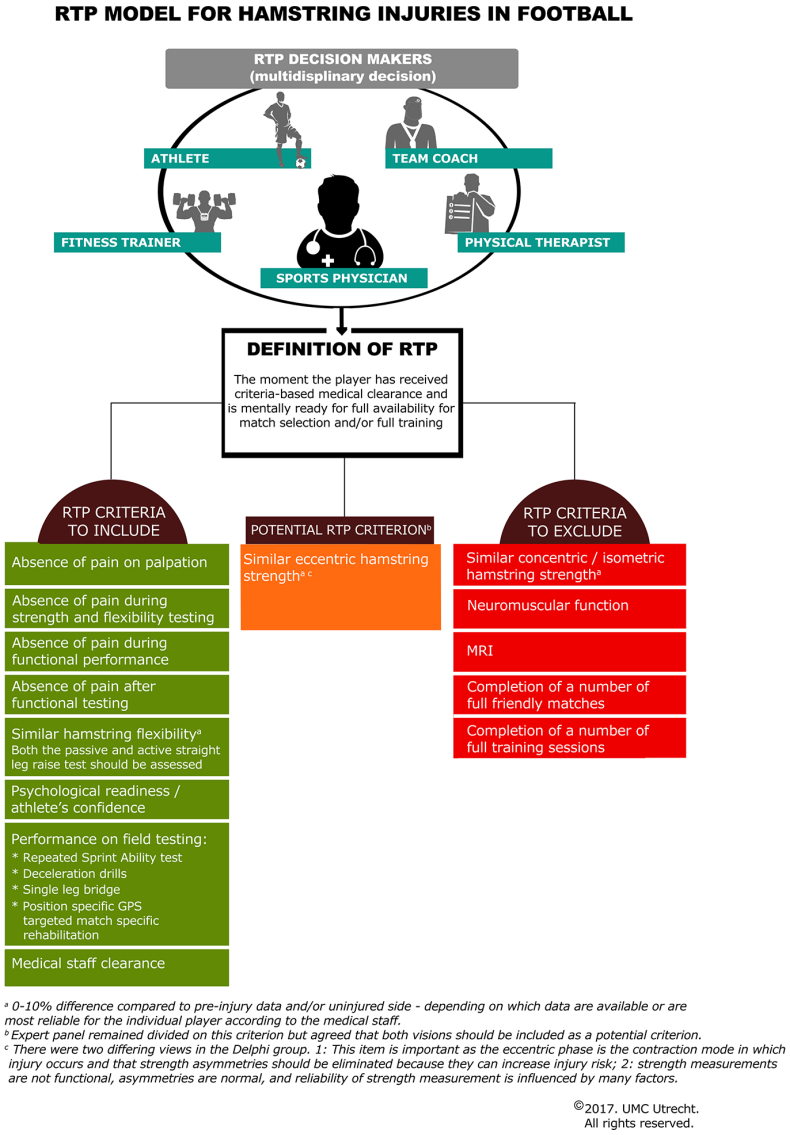

Figure 2 Return-to-play (RTP) model for hamstring injuries in football. GPS, global positioning system.

Lastly, the RTP model for hamstring injuries in football can assist clinicians with their decision-making when they decide on RTP in clinical practice.[5] This is particularly important in preventing hamstring injury recurrence due to too early RTP.

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